

Fluorine-Free Amphiphilic Copolymers for Broad-Spectrum Marine Biofouling Deterrence

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Biofouling, the accumulation of living organisms on submerged surfaces, remains a critical challenge in marine and industrial settings. Amphiphilic coatings offer promising antifouling performance by presenting both hydrophilic and hydrophobic domains, but often rely on environmentally persistent perfluoroalkyl substances (PFAS). This study introduces a fluorine-free amphiphilic copolymer synthesized via initiated Chemical Vapor Deposition (iCVD), incorporating 1,3,5,7-tetramethyl-1,3,5,7-tetravinyl cyclotetrasiloxane (V4D4) as the hydrophobic component and pyridinium-based zwitterions as the hydrophilic moiety. V4D4 enables an unprecedented conversion rate of pyridine to zwitterionic sulfobetaine, reaching 69.5% at a 240 nm depth into the film from a diffusion-limited vapor derivatization reaction, pointing to the network flexibility afforded by V4D4. The coating reduces *Pseudomonas aeruginosa* attachment by over 90% compared to aluminum surfaces and decreases *Cellulophaga lytica* biofilm formation by 55% compared to a zwitterionic benchmark under dynamic flow. Against the model diatom *Navicula incerta*, it achieves 99.7% removal after water jetting, outperforming a commercial fouling-release paint, Intersleek® 1100SR. The high fouling-release performance is attributed to its low Young's modulus, promoting deformation under shear. Additionally, the coating exhibits chemical stability in alkaline conditions that degrade PFAS-based counterparts. These findings present a durable, high-performance, and environmentally responsible alternative to PFAS-based amphiphilic copolymers as the next-generation antifouling technologies.

1. Introduction

Marine biofouling, the accumulation of bacteria, diatoms, and other marine organisms on submerged surfaces, presents a

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significant challenge across numerous industries, particularly in maritime transportation.^[1] The process of biofouling begins when microorganisms adhere to surfaces and form a biofilm, which subsequently attracts and supports the growth of more complex organisms, such as diatoms, barnacles, and mussels.^[2] The adhered organisms collectively increase drag, fuel consumption, emissions, and maintenance needs for ships and marine structures,^[3] while exacerbating the spread of invasive species, pointing to ecological imbalances in marine environments. The associated frictional resistance has been estimated to account for 70–90% of the total resistance for low-speed vessels;^[4] the biofouling-related expenses are believed to exceed \$56 million annually in the United States alone,^[5,6] highlighting the critical need for effective and sustainable antifouling strategies.

Over the years, a wide range of antifouling coatings have been developed to combat the detrimental effects of biofouling.^[7–11] Among these, amphiphilic polymer surfaces have emerged as particularly promising due to their ability to resist and release fouling by creating heterogeneous surfaces that are effective against a wide array of prokaryotic and eukaryotic

foulers.^[12,13] Amphiphilic coatings combine hydrophilic and hydrophobic domains within a single polymer chain, allowing them to present a dynamic surface that undergoes surface chain reorientation that further weakens the adhesion of organisms. Their antifouling and fouling-release performance has been shown to be effective across different marine environments.^[14]

Amphiphilic copolymer coatings can comprise block copolymers with segments bearing hydrophilic or hydrophobic side chains (e.g., synthesized using living polymerization techniques)^[15] or statistical copolymers with molecular heterogeneities. The latter often requires vapor-synthesis methods, like initiated Chemical Vapor Deposition (iCVD), to avoid microphase separation mediated by solvents and, hence, deliver high-quality conformal coatings with molecular heterogeneities.^[16,17] However, despite the concerns with perfluoroalkyl substances (PFAS), the amphiphilic copolymers reported to date have invariably relied on fluorine-containing

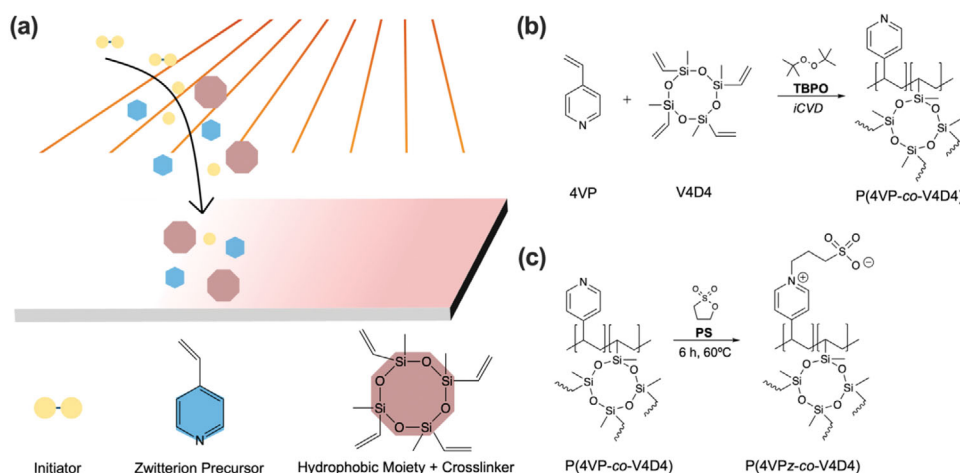


Figure 1. Synthesis scheme of the fluorine-free amphiphilic copolymers. a) Schematic of the iCVD technology for the one-step polymerization and thin film formation. Vaporized precursors are introduced into the vacuum reactor, where heterogeneous free-radical polymerization occurs. b) Reaction mechanism for the free-radical polymerization during iCVD. c) Reaction mechanism for subsequent reaction with 1,3-propane sultone (PS) to form zwitterionic moieties.

precursors for antifouling and fouling-release properties.^[18–20] Indeed, replacing those precursors with fluorine-free ones (e.g., siloxane) has led to amphiphilic copolymers with inferior biofouling-deterrence performance.^[21,22]

In addition, a fundamental challenge in the design of amphiphilic copolymers is to effectively bridge the different length scales at which biofouling occurs. While amphiphilic *block* copolymers deter eukaryotes (at length scales of 10 μm to 1 mm) effectively, *statistical* amphiphilic copolymers deter protein and bacteria (10 nm to 10 μm), leaving an unmet need in the simultaneous deterrence of foulers at the two length scales. Such broad-spectrum biofouling deterrence is crucial for an effective and long-lasting antifouling solution, particularly given the complex taxonomic diversity of marine biofouling.^[23]

To address these two bottlenecks, namely the reliance on PFAS-based materials and the unmet need for fouling deterrence across length scales, this work designs a new class of amphiphilic copolymers that present statistical amphiphilicity without relying on fluorine while enabling elasticity to release eukaryotic foulers (Figure 1).^[13] The environmentally friendly amphiphilic coatings avoid PFAS by replacing fluorine-bearing precursors with a tetrafunctional crosslinker, 1,3,5,7-tetramethyl-1,3,5,7-tetravinyl cyclo-tetrasiloxane (V4D4).^[24,25] Unlike PFAS, V4D4 does not pose long-term environmental persistence or toxicity concerns, making it a more sustainable hydrophobic alternative for amphiphilic coatings. That choice of precursor is further motivated by the excellent chemical resistance, thermal stability, and surface hydrophobicity of organosilicon compounds.^[26]

To achieve statistical amphiphilicity, V4D4 was copolymerized with 4-vinylpyridinium-based sulfobetaine to form P(4-vinylpyridine (4VP)-co-V4D4) (P(4VPz-co-V4D4)) using an all-dry polymer synthesis and processing method, namely iCVD. iCVD was chosen because it enables the one-step copolymerization of hydrophilic and hydrophobic monomers and the direct deposition of amphiphilic polymers with nanoscale heterogeneities, which is challenging for solution-based processes (as

they are prone to micro-phase separation).^[17] Serendipitously, the incorporation of V4D4 was discovered to lead to a high content of zwitterionic moieties, which were obtained via a vapor-phase derivatization reaction converting a pyridine side group to the pyridinium sulfobetaine, hinting at the high permeability of the amphiphilic thin films.

To evaluate the performance of P(4VPz-co-V4D4), two well-established polymer systems were used as benchmarks. P(4VPz-co-1H,1H,2H,2H-perfluorooctyl acrylate) (P(4VPz-co-PFOA)) has been recognized as a high-performance amphiphilic copolymer featuring PFAS-based perfluoroalkyl side chains that enable ultra-low surface energy and dynamic chain reorientation;^[20] however, its fouling-release properties have not been formally assessed. P(4VPz-co-divinylbenzene) (P(4VPz-co-DVB)) is a widely used zwitterionic polymer with a crosslinked network and reduced chain mobility,^[27] serving as a benchmark for mechanical stability and antifouling performance of zwitterionic polymers. By comparing P(4VPz-co-V4D4) against these antifouling coatings demonstrated in the literature, this study assessed whether fluorine-free amphiphilic coatings can achieve fouling deterrence while eliminating PFAS use and maintaining the chemical stability of crosslinked coatings.

This new class of amphiphilic copolymers also addresses a critical gap in antifouling materials design by providing simultaneous resistance to prokaryotic and eukaryotic foulers across different length scales. At the air-liquid-solid interface, the coatings effectively deter biofilm formation by *Pseudomonas aeruginosa*, a widely studied model bacterium in biofilm studies,^[28] which is also present in marine environments.^[29] Under simulated marine conditions, the coatings deter *Cellulophaga lytica*, a marine bacterium widely used in biofouling studies.^[19] Its antifouling and fouling release efficacy against *Navicula incerta*, a marine diatom frequently used for marine fouling studies,^[30] is also confirmed. We attributed the broad-spectrum antifouling performance to a unique combination of (i) molecular heterogeneity, which deters microscopic foulers like bacteria, and

(ii) mechanical elasticity, which allows for dynamic adaptation of the interface under mechanical stress, facilitating the fouling release of marine eukaryotes. This attribution is substantiated by the excellent surface chain reorientation of the amphiphilic copolymer thin films, characterized via dynamic water contact angle measurements and their interfacial elasticity, demonstrated using atomic force spectroscopy (AFM).

Beyond its antifouling performance, P(4VPz-co-V4D4) exhibited enhanced chemical stability compared to PFAS-based amphiphilic coatings, ensuring long-term durability. By eliminating the reliance on environmentally persistent PFAS while achieving comparable or superior antifouling and fouling release efficacy across different domains of foulers, this work presents a next-generation solution for sustainable antifouling coatings. The impact of these findings extends beyond marine antifouling to medical implants,^[31] membrane separation,^[32] and tissue engineering.^[33]

2. Results and Discussion

2.1. The All-Dry Synthesis of Amphiphilic Copolymers and a Complete Derivatization to Yield Zwitterion-Based Amphiphilicity

We synthesized the statistical amphiphilic copolymer using a two-step procedure: iCVD polymerization was used to make a precursor film comprising 4VP and V4D4 with varying compositions (Figure 1a,b), which subsequently underwent a derivatization reaction with 1,3-propanesultone (PS) (Figure 1c) to convert the pyridine moieties to pyridinium-based sulfobetaine zwitterions. The iCVD process was carried out under medium vacuum conditions of <1 Torr. Monomer vapors, along with inert gas and initiator flows, were metered into the reactor chamber, where they physisorbed onto the cooled substrate (kept at 36 °C). A heated filament array, composed of nickel/chromium, initiated the decomposition of *tert*-butyl peroxide (TBPO), generating *tert*-butoxy radicals. These radicals chemisorbed onto the surface-adsorbed monomers, triggering free-radical polymerization directly on the substrate. The coating thickness was monitored in situ in real time, and the polymerization process was halted once the desired thickness (100–200 nm) was achieved.

The successful synthesis of P(4VPz-co-V4D4) copolymers of different compositions was confirmed using Fourier transform infrared (FTIR) and X-ray photoelectron spectroscopy (XPS). The FTIR spectra (baseline-corrected and normalized by the film thickness) of homopolymers P4VP and PV4D4 and their copolymers of varying compositions are shown in Figure 2a. One of the characteristic peaks of V4D4 was the Si-O-Si asymmetric stretching of the cyclotetrasiloxane ring at 1058 cm⁻¹. The peak at 1260 cm⁻¹ corresponds to the Si-CH₃ symmetric bending. Peaks at 1597 and 1415 cm⁻¹ corresponded to the vibration of the pyridine ring in 4VP, and the broad absorption at around 3000 cm⁻¹ corresponded to alkyl C-H bonds. These peaks confirm the presence of both 4VP and V4D4 in the copolymer thin films. Furthermore, the subtle shift in the Si-O-Si peak at 1058 cm⁻¹, indicating that the siloxane bond is influenced by the incorporation of 4VP, serves as evidence that the components are chemically bonded as a copolymer (instead of a physical mixture of homopolymers). Using the peaks at 1597 cm⁻¹ and 1058 cm⁻¹, we calculated the

copolymer film compositions to have molar fractions of V4D4 (F_{V4D4}) of 31%, 45%, and 67%.

After iCVD polymerization, PS derivatization was performed on the as-synthesized homopolymer and copolymer thin films. 4VP has been used to produce zwitterionic moieties via vapor-phase derivatization through exposure to PS vapor, giving rise to zwitterionic sulfobetaine functional moieties via a ring-opening reaction.^[34] The superhydrophilicity of the zwitterionic groups ensures a drastic contrast in surface energy between the hydrophilic and hydrophobic moieties in the amphiphilic chemistry, which has been paired with fluorinated side chains to deter *P. aeruginosa* biofilms at the air-liquid-solid surface.^[18]

The successful derivatization of P4VP and P(4VPz-co-V4D4) was supported by FTIR (Figure 2c), as confirmed by the peak at 1036 cm⁻¹ (dashed line) in all spectra, corresponding to the S-O stretching in sulfobetaine. The derived polymer will be denoted as P(4VPz-co-V4D4). Interestingly, the broad peaks at around 1200 cm⁻¹, corresponding to sulfonate salts ($-\text{SO}_3^-$, red shaded), were more prominent in P(4VPz-co-V4D4) copolymers than the homopolymer P4VPz, indicating nearly complete conversion of pyridine to pyridinium (as these spectra reflect bulk compositions based on the transmission mode employed in FTIR). That high conversion was further evidenced by the shift of the pyridine peak from 1597 to 1640 cm⁻¹ (dashed lines in Figure 2c) across all P(4VPz-co-V4D4) with varying V4D4 compositions, corresponding to the formation of pyridinium groups.^[35]

The observed peak shift, indicating complete conversion from pyridine to pyridinium, is unprecedented for similar derivatization reactions. Provided that previous research has shown that lightly crosslinked P(4VPz-co-DVB) has a penetration depth of several nanometers, limiting the conversion to the surface.^[36] In contrast, the deeper penetration observed here represents a distinct outcome of incorporating V4D4 as a crosslinker. This deep derivatization depth was further validated by XPS depth profiling of P(4VPz-co-V4D4) (with F_{V4D4} of 67%). In XPS, successful derivatization is indicated by the high-resolution N(1s) scan before and after derivatization (Figure 2b), where 399 eV and 402 eV correspond to unreacted pyridine and derivatized quaternary pyridinium, respectively. The complete peak shift upon derivatization proved the full conversion. Figure S1a (Supporting Information) showed high-resolution N(1s) spectra measured at incremental depths from 0 to 240 nm into the copolymer film, achieved by Ar⁺ cluster etching. At the surface, the conversion rate of pyridine to pyridinium was ≈84%, and even at a depth of 240 nm, the conversion rate remained at ≈70% (Figure S1b, Supporting Information). Furthermore, the atomic ratio of sulfur (S) to nitrogen (N), indicative of sulfobetaine group distribution, remained close to one throughout the film depths, corroborating the complete derivatization. The penetration depth far exceeded previously reported values, which were on the order of several nanometers. We attributed that to the cyclic siloxane structure in V4D4, which likely led to larger and greater numbers of free-volume elements throughout the copolymer films than uncrosslinked films or those using crosslinkers like DVB. In addition, the flexibility of the Si-O-Si bond leads to greater chain mobility and, hence, diffusivity, enabling a deeper penetration during the derivatization reaction. This theory also aligns with the Young's modulus we quantified of the samples, which will be discussed in the following section.

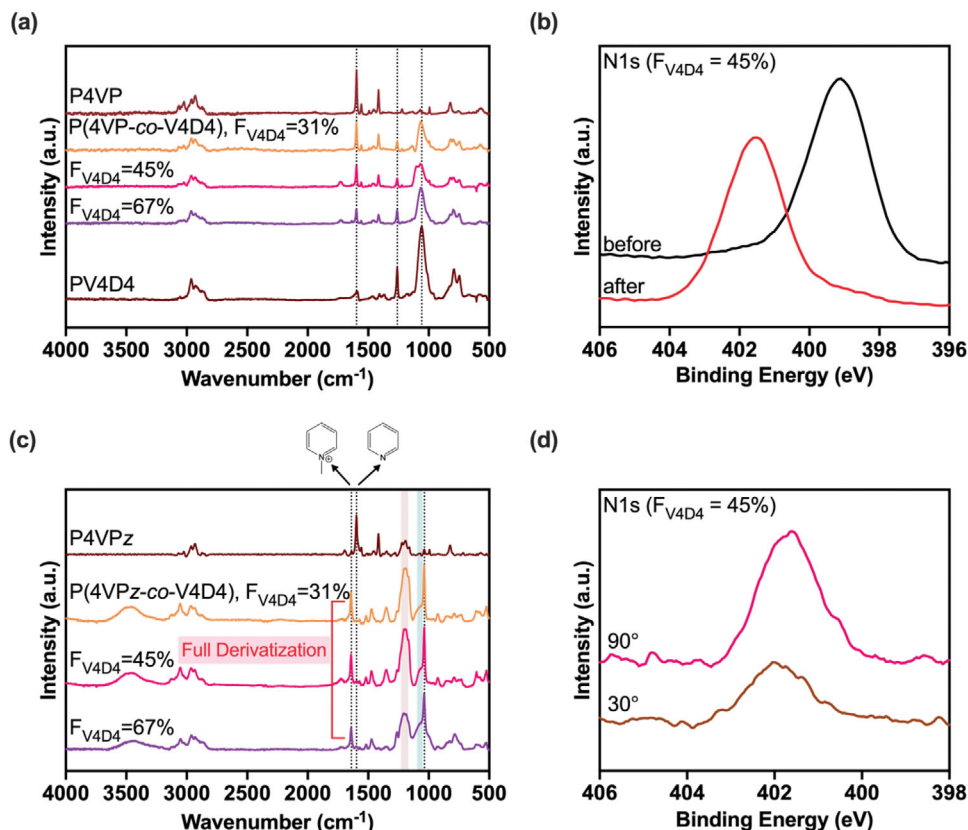


Figure 2. Chemical characterization of the copolymer thin films. a) FTIR of homopolymer P4VP and PV4D4, and P(4VP-co-V4D4) copolymers with V4D4 molar fractions (F_{V4D4}) of 31%, 45%, and 67%. The characteristic absorptions are marked with dashed lines, corresponding to the pyridine ring (1597 cm^{-1}), symmetric Si-CH₃ bending (1260 cm^{-1}), and asymmetric Si-O-Si vibration (1058 cm^{-1}). b) High-resolution XPS scans of N(1s) for P(4VP-co-V4D4) with F_{V4D4} of 45% before and after derivatization with PS. c) FTIR of derivatized homopolymer P4VPz and copolymers P(4VPz-co-V4D4) with F_{V4D4} of 31%, 45%, and 67%. The peak at 1036 cm^{-1} (dashed line) corresponds to the S-O stretching in sulfobetaine and indicates successful derivatization. The peaks at 1597 cm^{-1} and 1640 cm^{-1} (dashed lines) correspond to pyridine and converted pyridinium, respectively. The asymmetric Si-O-Si vibration is highlighted in blue (around 1056 cm^{-1}), and sulfonate (SO_3^-) highlighted in red (around 1200 cm^{-1}). d) AR-XPS high-resolution N(1s) scan of P(4VPz-co-V4D4) with F_{V4D4} of 45% collected at angles of 30° and 90° .

Elemental mapping results obtained using AR-XPS (measured at angles of 30° and 90°) are shown in Table S1 (Supporting Information). The simultaneous presence of nitrogen (from 4VP) and silicon (from V4D4) at both angles confirmed successful copolymerization. The near-surface molar ratio of V4D4 to 4VP was calculated using the atomic percentage of silicon relative to nitrogen to avoid the influence of adventitious carbon and oxygen. Interestingly, the calculated molar ratio of V4D4 to 4VP (for an F_{V4D4} of 45%) measured at 90° was 0.8, and that measured at 30° was 1.1, indicating the preferential presence of V4D4 near the film top surface. That near-surface preference for V4D4 persisted after the derivatization reaction, as shown by the survey scan and AR-XPS high-resolution scans on N(1s) performed on P(4VPz-co-V4D4) (Figure 2d). We attributed the spatial distribution of V4D4 to its lower surface energy (20.6 dyne cm^{-1}) compared to 4VP (36.2 dyne cm^{-1}).^[37,38] Further validation of this preferential distribution was provided by XPS depth profiling of P(4VPz-co-V4D4), as shown in Figure S1c (Supporting Information). The V4D4-to-4VP ratio was the highest at the surface and remained at a lower value deeper into the film.

Raman spectroscopy, known for detecting changes and defects in siloxane ring structures that could occur during CVD processes,^[39,40] was used to assess the retainment of the siloxane ring in V4D4. As shown in Figure S2 (Supporting Information), the characteristic peak of the 4-membered siloxane ring in V4D4 at 475 cm^{-1} was retained in the as-synthesized P(4VPz-co-V4D4) copolymer (with F_{V4D4} of 45%). This confirmed that the integrity of the siloxane ring structure was preserved after both the iCVD and PS derivatization processes. This work marks the first successful synthesis of a fully converted zwitterionic polymer film via a vapor-phase technique, pointing to improved stability of the antifouling performance (e.g., in case of film degradation during prolonged use).

2.2. Enhanced Chemical Stability of the Amphiphilic Copolymers Compared to PFAS-Based Counterparts

Chemical stability is a critical factor in the performance and longevity of antifouling coatings, particularly with regard to reducing PFAS release into the environment. The amphiphilic

copolymers were designed to reduce PFAS release into the environment. In this context, the coating stability testing in acidic and alkaline solutions is an important metric, which has been used to illicit PFAS release from fluorine-containing polymers in prior studies.^[41,42]

To ensure that our coatings can withstand challenging conditions, we rigorously assessed the chemical stability of PV4D4 in acidic and alkaline environments by submerging them in 0.5 M H₂SO₄ or 1 M NaOH solution for 5 h. We then compared its chemical stability to that of the hydrophobic moiety used in previous generations of statistical amphiphilic copolymers, i.e., PFOA.

FTIR results of untreated PPFOA coatings, compared to those treated with acid and base, were shown in Figure S3b (Supporting Information). While PPFOA remained chemically intact in acidic environments, its integrity was significantly compromised after exposure to alkaline conditions. The peak at 1738 cm⁻¹ corresponds to the stretching of C=O in PFOA, the peaks at 1200 cm⁻¹ correspond to -CF₂- stretching, and the peaks at 1230 cm⁻¹ correspond to -CF₂-CF₃ end group. After the alkaline treatment, the area under the 1738 cm⁻¹ peak increased significantly relative to the peaks at 1200 and 1230 cm⁻¹, indicating a substantial loss of the fluorocarbon side chains. This was likely due to the S_N2 hydrolysis that we have demonstrated previously,^[41] which has also been commonly observed in long-chain PFAS compounds.^[43] The chemical degradation not only releases harmful PFAS to the environment, but also undermines the antifouling performance of the statistical amphiphilic materials.

In contrast, the FTIR results of untreated PV4D4 coatings versus those after acid and base treatment, shown in Figure S3a (Supporting Information), revealed no change in the coating composition. It confirmed the excellent chemical stability of PV4D4. Moreover, compared to PFOA, V4D4 also functions as a crosslinker, further ensuring the structural and chemical integrity of the amphiphilic copolymer. These results confirmed V4D4 as a promising replacement for fluorinated moieties in the design of statistical amphiphilic copolymers, conferring hydrophobicity and stability while avoiding PFAS entirely. While assessment of the coating durability in practical applications is beyond the scope of this report, we note the use of similar procedures in the literature as accelerated degradation tests. Prior studies have demonstrated that chemical stability tests performed under acidic or alkaline conditions effectively predict the long-term material degradation behavior and enable rapid assessment of coating durability.^[44,45] As such, we predict that the amphiphilic coatings reported here have promising long-term stability, but future studies that involve long-term exposure would more accurately assess the durability of the coatings.

While the long-term stability of P(4VPz-co-V4D4) is not directly assessed in this report, such stability has been well established in the literature for iCVD-synthesized cyclosiloxane polymers in general.^[46,47] For example, poly(trivinyltrimethylcyclotrisiloxane) (poly(V3D3)) remained hydrolytically and electrically stable for over two years under physiological soak conditions and continuous electrical bias.^[48] The stability of the PV3D3 thin film impedance speaks to its pinhole-free nature, which also indicates that the film morphology remains unchanged over prolonged immersion conditions. In another study, PV4D4-coated surfaces maintained high proliferative capacity when exposed to culture of human mes-

enchymal stem cells for up to 11 days, pointing to its interfacial stability.^[49] Furthermore, a biocompatible polymer composite incorporating a V4D4 adhesion-promoting layer retained > 96% antibacterial activity against MRSA after 9 weeks of immersion in phosphate-buffered saline; the absence of delamination or other types of coating damage demonstrates long-term chemical and morphological stability of V4D4-based thin films.^[50] Collectively, these reports point to the exceptional long-term stability expected of iCVD-synthesized siloxane copolymers, like the one reported in this study.

Additional experiments were performed to examine the toxicity of potential degradation products from the coating. We used *C. lytica* as the model strain in these experiments. Briefly, a 0.1 M NaOH solution was used to extract potential degradation products from P(4VPz-co-V4D4) and an aluminum surface (as a reference, see Materials and Methods section for details). After neutralization, the extraction solutions were incubated with *C. lytica* for 5 h, the CFU counts of which were compared (Figure S4, Supporting Information). The exposure led to no statistically significant difference in the bacterial viability, suggesting that P(4VPz-co-V4D4) did not produce any compounds that exhibit acute toxicity to marine bacteria. This is in stark contrast to the widely reported toxicity of the hydrolysis products of fluorine-containing polymers.^[51,52] Nevertheless, future work on the comprehensive assessment of the ecotoxicity of the different classes of amphiphilic coatings, which often examines the growth inhibition against marine animals,^[53] is required to rigorously compare their environmental effects.

2.3. Ultra-Smooth and Elastic Amphiphilic Copolymer Thin Films

Surface roughness plays a crucial role in the performance of antifouling polymer thin films.^[54] A lower roughness is often desirable for thin film coating, which has been associated with reduced bacterial adhesion and biofilm formation.^[34,55] To characterize the surface roughness of the copolymer thin films, AFM was employed on coated Si wafers (Figure 3a–e). The successful synthesis P(4VPz-co-PFOA) was confirmed with FTIR results in Figure S5 (Supporting Information). The height mapping of P(4VPz-co-V4D4) with varying V4D4 compositions, P(4VPz-co-DVB), and P(4VPz-co-PFOA) revealed that all the copolymer thin films were extremely smooth and pinhole-free, with root mean square roughness (R_{rms}) less than 0.6 nm. The variance of smoothness was minimal across V4D4 compositions (i.e., 67%, 45%, and 31%). This ultra-smoothness is comparable to other amphiphilic copolymers produced via iCVD.^[56] This supports the molecular heterogeneity of iCVD-synthesized amphiphilic copolymer thin films, as phase-separated microdomains (which are common among solution-synthesized films)^[57] were absent due to the adsorption-driven vapor-based deposition process.^[58]

We also leveraged AFM for nanomechanical mapping, which quantifies the Young's modulus of the thin films. Elastic coatings can dislodge fouling organisms via deformation when the surface is subjected to shear forces, such as during ship cruising.^[59] As such, coatings with lower Young's moduli tend to have better fouling-release characteristics. Figure 3e summarizes the modulus distribution based on the areal modulus mapping results for P(4VPz-co-V4D4) with varying V4D4 compositions, as well as for

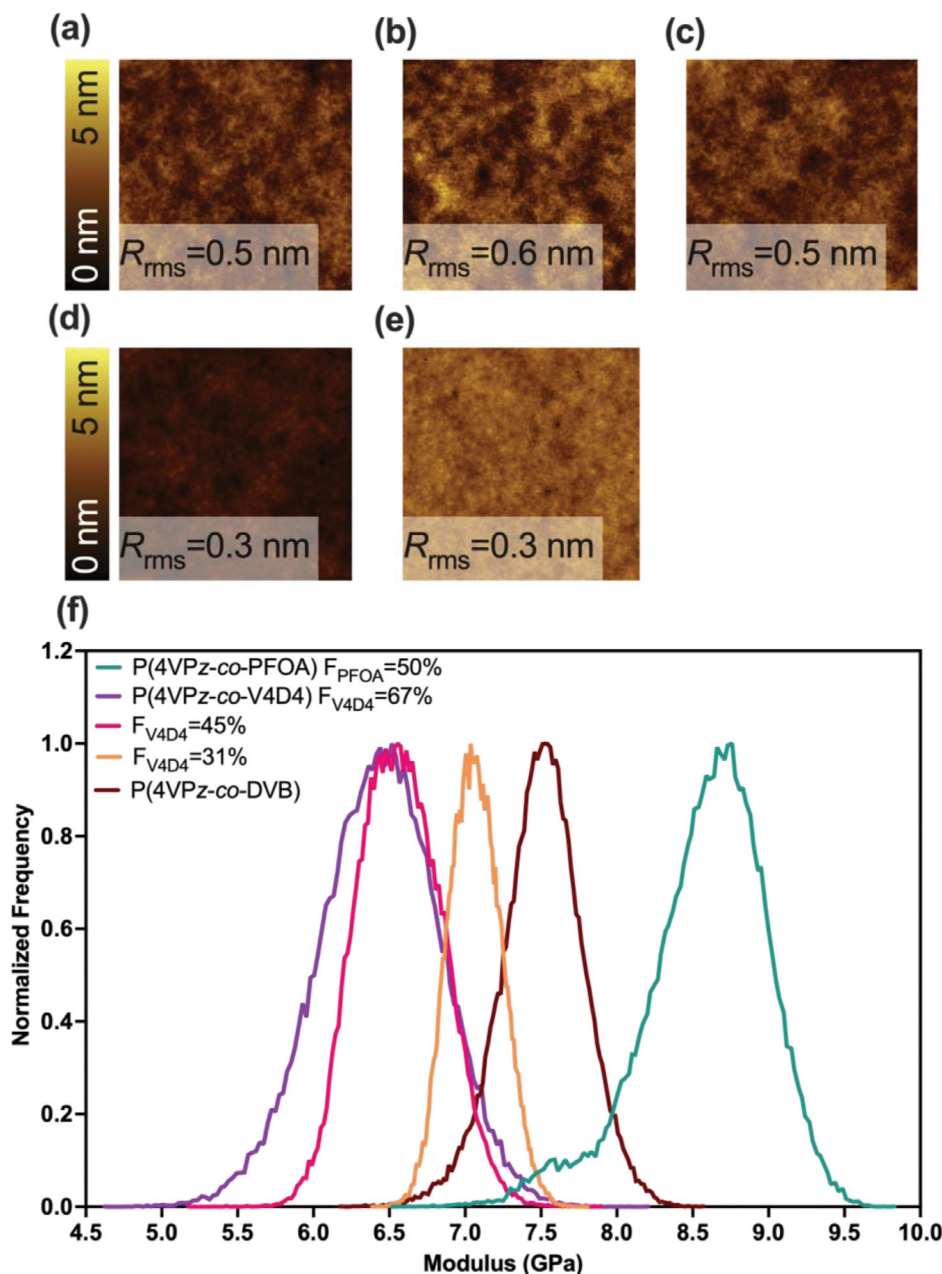


Figure 3. Surface roughness and Young's modulus of the copolymer thin films. AFM height images of a $5 \times 5 \mu\text{m}^2$ area of iCVD coatings of P(4VPz-co-V4D4) with F_{V4D4} of a) 67%, b) 45%, c) 31%, d) P(4VPz-co-DVB), and e) P(4VPz-co-PFOA). The RMS surface roughness is shown in each image. f) Distribution of the Young's moduli measured for each copolymer composition. $n = 3$.

P(4VPz-co-DVB) and P(4VPz-co-PFOA). The peaks in the graph correspond to the most representative modulus values for each sample. The results showed that P(4VPz-co-V4D4) with F_{V4D4} of 45% and 67% exhibited the lowest moduli, measuring 6.4 GPa (95% confidence level (CI): 6.2 – 6.6) and 6.5 GPa (95% CI: 6.4 – 6.6), respectively. The modulus increased to 7.1 GPa (95% CI: 7.0 – 7.2) for P(4VPz-co-V4D4), $F_{\text{V4D4}} = 31\%$, while P(4VPz-co-DVB) had a modulus of 7.4 GPa (95% CI: 7.3 – 7.5). P(4VPz-co-PFOA) exhibited the highest modulus of 8.0 GPa (95% CI: 7.8 – 8.2).

This trend suggests that increasing V4D4 content results in copolymers with lower moduli. Additionally, V4D4-crosslinked

copolymers exhibit lower moduli than both DVB-crosslinked and PFOA-based copolymers. This reduction in modulus can be attributed to (i) the siloxane bonds being inherently flexible, allowing for greater chain mobility, and (ii) the cyclic siloxane structure of V4D4 potentially creating free-volume elements within the polymer matrix. Furthermore, despite possessing multiple vinyl groups, V4D4 has been associated with a relatively moderate reactivity, leading to a lower crosslinking density compared to other crosslinkers at similar mole fractions, thereby contributing to a reduced network rigidity.

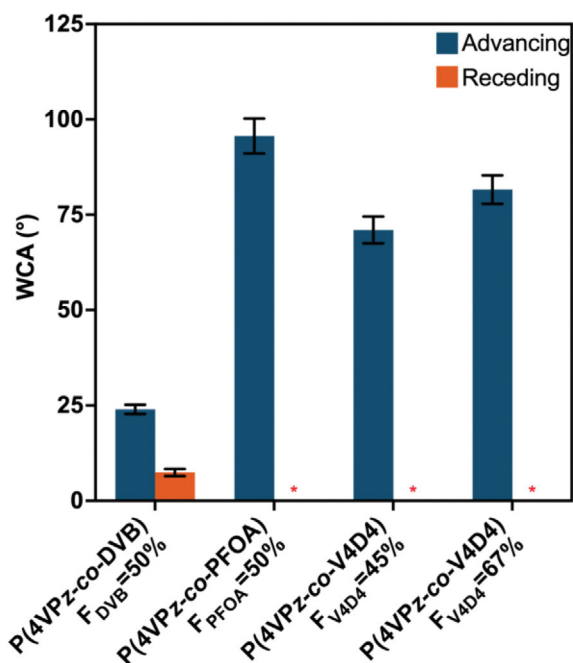


Figure 4. Advancing and receding water contact angle (WCA) measurements for P(4VPz-co-DVB) with F_{DVB} of 50%, P(4VPz-co-PFOA) with F_{PFOA} of 50%, P(4VPz-co-V4D4) copolymers with F_{V4D4} of 45% and 67%. The asterisk indicates that the receding WCA for the corresponding surfaces was close to zero.

2.4. Uncompromised Dynamic Wetting and Surface Chain Reorientation in the Fluorine-Free Amphiphilic Copolymers

We measured the dynamic water contact angle (WCA) to characterize the dynamic wetting behavior of the amphiphilic copolymers. Previous research has shown that statistical amphiphilic copolymers resist biofilm formation via facile surface chain reorientation at the air-liquid-solid interface,^[60] presenting hydrophobic side chains when dry and hydrophilic side chains when wet. As such, the hydrophilic side chains reduce the adhesion of foulants at the liquid-solid interface (e.g., when the surface is submerged), whereas the hydrophobic side chains promote self-cleaning properties at the air-solid interface (e.g., when the surface is exposed to air). Dynamic WCA measurements differentiate the surface energy under wet and dry conditions, providing insights into its dynamic hydration.

Dynamic WCA results of P(4VPz-co-V4D4) with F_{V4D4} of 45% and 67%, and their PFAS-based counterpart, P(4VPz-co-PFOA) with F_{PFOA} of 50% are illustrated in Figure 4. The benchmark, P(4VPz-co-PFOA), which is well-documented for its facile chain reorientation, exhibited an advancing WCA of $96 \pm 5^\circ$ and a close-to-zero receding WCA with $F_{PFOA} = 50\%$. P(4VPz-co-V4D4) with $F_{V4D4} = 45\%$ showed an advancing WCA of $71 \pm 4^\circ$ and a close-to-zero receding WCA (which is not quantified due to the loss of contact between the dispensing syringe and the droplet). P(4VPz-co-V4D4) with $F_{V4D4} = 67\%$ showed an advancing WCA of $82 \pm 4^\circ$ and a close-to-zero receding WCA. In both cases, the large hysteresis of over 70° hints at the facile surface chain reorientation, with the hydrophobic V4D4 exposed when dry but buried upon wetting to minimize interfacial energy, while the zwitteri-

onic moieties reorient rapidly to the topmost surface upon wetting. The increase in advancing WCA as F_{V4D4} increases from 45% to 67% indicates the hydrophobicity of V4D4, and the close-to-zero receding WCA confirms the superhydrophilicity of the zwitterionic moieties. The comparable WCA hysteresis among P(4VPz-co-PFOA) and P(4VPz-co-V4D4) copolymers highlights the ability of V4D4, as a fluorine-free alternative, to enable dynamic behaviors while eliminating PFAS.

Furthermore, to enable a direct comparison of the effects of different crosslinkers on the dynamic wetting behavior, we chose a commonly used crosslinker, DVB, as the benchmark.^[61] We synthesized a copolymer made of DVB and 4VP, with a DVB molar fraction of $\approx 50\%$, which was subjected to the vapor-phase derivatization reaction to yield a zwitterion-based amphiphilic copolymer that we denoted as P(4VPz-co-DVB), $F_{DVB} = 50\%$. Figure 4 also shows the advancing and receding WCA for P(4VPz-co-DVB), $F_{DVB} = 50\%$. In contrast, P(4VPz-co-DVB) with F_{DVB} of 50% had an advancing WCA of $24 \pm 1^\circ$ and a receding WCA of $7 \pm 1^\circ$, corresponding to a hysteresis of 17° . Compared to DVB-based copolymers, the larger hysteresis observed in the V4D4-based copolymers is attributed to the increased free volume and chain mobility provided by V4D4's cyclic siloxane structure, enabling enhanced dynamic chain reorientation. This reorientation ability has been shown to be crucial in antifouling performance, particularly against *P. aeruginosa* at the air-liquid-solid interface. Furthermore, compared to DVB-based copolymers, V4D4-based amphiphilic copolymer significantly improves the antifouling performance against marine bacteria *C. lytica* under marine dynamic conditions, as will be introduced in the following section. This improvement is likely resulted from the superior dynamic chain reorientation ability of V4D4-based copolymers, ensuring maximum exposure of hydrophilic moieties underwater.

Based on the surface wetting properties and the biofouling results, it is tempting to explain the antifouling performance of the amphiphilic material, P(4VPz-co-V4D4), in the context of the Baier curve, a commonly used framework to characterize antifouling materials. However, we note that the Baier curve does not apply in this case due to the highly dynamic nature of the copolymer with molecular heterogeneities. The large WCA hysteresis places the advancing and receding WCA of the amphiphilic copolymer in entirely different regimes along the Baier curve, highlighting its dynamic nature. While not within the scope of the current report, our future research will focus on the development of new theoretical frameworks to describe this new category of dynamic antifouling materials.

2.5. Simultaneous Antifouling and Fouling-Release Capabilities of the Fluorine-Free Amphiphilic Copolymers

To assess the antifouling performance of the fluorine-free amphiphilic copolymers, we chose to incubate the coated surfaces with concentrated cultures of three model foulers: *P. aeruginosa* (PAO1 strain) for bacterial biofilm at the solid-air-liquid three-phase interface, *C. lytica* as a model strain for marine bacteria biofilm, and diatom *N. incerta* as a marine eukaryotic fouler. We note that such tests have been shown to be more effective at inducing biofouling than prolonged exposure to natural seawater, as illustrated in previous studies.^[27] Aluminum was used as a

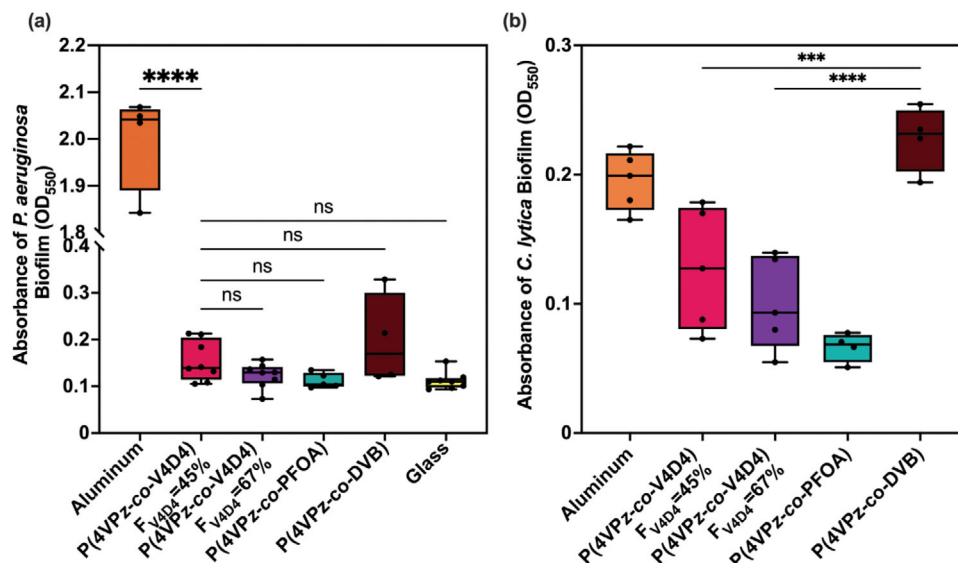


Figure 5. Evaluation of the antifouling performance using two model bacteria in box plots, where boxes indicate the interquartile range (middle 50% of data) with the line marking the median, and “whiskers” extended from boxes show the minimum and maximum values. a) Absorbance of the CV staining applied to *P. aeruginosa* biofilms (OD₅₅₀) grown at the solid-air-liquid interface on a control aluminum surface (with the native oxide), P(4VPz-co-V4D4) copolymers with F_{V4D4} of 67% and 45%, P(4VPz-co-PFOA), P(4VPz-co-DVB), and glass coverslips for 8 h under static incubation conditions. b) Absorbance of the CV staining applied to *C. lytica* biofilms (OD₅₅₀) cultured on submerged surfaces of control aluminum, P(4VPz-co-DVB), P(4VPz-co-PFOA), and P(4VPz-co-V4D4) copolymers with F_{V4D4} of 67% and 45% for 24 h in a flow cell. $n \geq 4$.

common reference material for the comparison of antifouling performance across different fouling species. It is chosen based on its widespread use in marine applications, particularly in ship hulls and offshore structures, where biofouling is a persistent challenge.^[62]

P. aeruginosa was selected as a representative biofilm-forming bacterium due to its importance in both biomedical and marine environments. It is an opportunistic nosocomial pathogen frequently found on medical devices, surgical instruments, and glassware.^[28] Additionally, as an opportunistic marine bacterium, *P. aeruginosa* can contribute to biofouling while causing health concerns.^[29] More importantly, *P. aeruginosa* is known to form stable and reproducible biofilms at the air-liquid-solid interface under static culture conditions, which has been considered an important performance metric in amphiphilic materials design. The reduced biofilm formation at the three-phase interface has been associated with the fouling-deterrence mechanism of facile surface chain reorientation.^[20] To assess the antifouling performance of polymer coatings using *P. aeruginosa* as a model fouler, glass was chosen as the negative reference material. It is commonly used in the literature, thus enabling direct benchmarking of the results reported here.

Biofilms formed at the solid-air-liquid interface by *P. aeruginosa* after 8 h of static incubation were quantified using the crystal violet (CV) staining (Figure 5a). In addition to aluminum and glass, P(4VPz-co-PFOA) and P(4VPz-co-DVB) were used as benchmarks based on their well-established antifouling properties.^[17] No statistically significant differences were observed among the biofilms formed on P(4VPz-co-V4D4) with F_{V4D4} of 67% and 45%, P(4VPz-co-PFOA), P(4VPz-co-DVB), and glass coverslips. Notably, *P. aeruginosa* biofilm formation on the previously established best-in-class amphiphilic copolymer,

P(4VPz-co-PFOA), was comparable to that on P(4VPz-co-V4D4), with no statistically significant differences. This comparable antifouling performance is attributed to the retained chain reorientation ability in the V4D4-based copolymers, compared to that of the PFOA-based ones. Importantly, the incorporation of V4D4 not only replaces the fluorinated moiety but also enhances the accessibility and conversion of pyridine groups into zwitterionic moieties as previously discussed. This high degree of zwitterionic functionalization promotes a more hydrated and dynamic interface under aqueous conditions, which plays a critical role in deterring biofilm formation. All amphiphilic and zwitterionic coatings demonstrated strong antifouling performances, reducing *P. aeruginosa* biofilm formation by at least 90% compared to aluminum.

C. lytica, a marine bacterium commonly found in biofilms on ship hulls, serves as a model foulant for many marine antifouling studies.^[63] It is essential that *C. lytica* is cultured under flow to recapitulate the dynamic marine environments in which its biofilm formation occurs. Furthermore, the fully submerged culture of *C. lytica* provides insights into the importance of surface hydration as an antifouling mechanism in the materials design. To reflect this environment, *C. lytica* biofilms were grown under flow conditions for 24 hours using a custom-built flow cell system (Figure S6, Supporting Information), with aluminum used as the reference substrate due to its relevance in ship hull construction.^[64]

The P(4VPz-co-V4D4) amphiphilic copolymers outperformed P(4VPz-co-DVB) and aluminum (Figure 5b). Notably, the best-performing amphiphilic copolymer, P(4VPz-co-V4D4) with F_{V4D4} = 67%, reduced *C. lytica* biofilm by 55% compared to its zwitterionic counterpart, P(4VPz-co-DVB) ($p = 0.0006$); and the other amphiphilic copolymer composition with F_{V4D4} = 45% also reduced *C. lytica* biofilm by 44% compared to

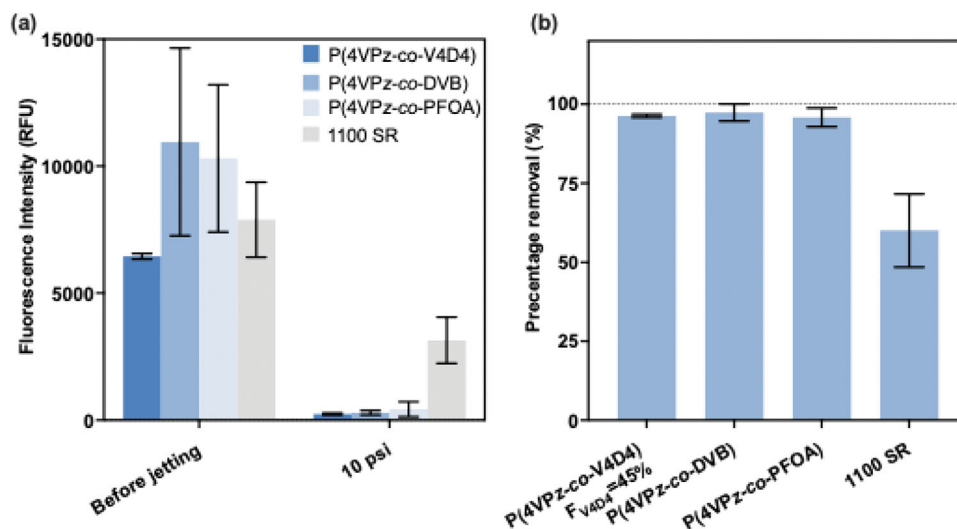


Figure 6. Evaluation of the antifouling and fouling release performance of P(4VPz-co-V4D4), $F_{V4D4} = 45\%$, P(4VPz-co-DVB), and P(4VPz-co-PFOA) against marine diatom *N. incerta* compared to commercial fouling release paint Intersleek 1100SR: a) Fluorescence intensity of chlorophyll extracted from surface-attached diatoms after 2 h of incubation (before jetting), and from the remaining diatoms after water-jet removal at 10 psi. b) Calculated percentage removal after waterjetting at 10 psi.

P(4VPz-co-DVB) ($p = 0.0156$). This enhanced antifouling performance is attributed to the superior chain reorientation ability of P(4VPz-co-V4D4), as evidenced by the WCA measurements, which facilitates fouling release. Additionally, the larger contrast in surface free energy between the hydrophilic and hydrophobic domains in P(4VPz-co-V4D4) compared to P(4VPz-co-DVB) likely further contributes to its improved antifouling efficacy. There were no statistically significant differences in antifouling performance between P(4VPz-co-V4D4) with $F_{V4D4} = 67\%$ and $F_{V4D4} = 45\%$. Moreover, a comparison to the previous best-in-class amphiphilic copolymer, P(4VPz-co-PFOA), revealed that the fluorine-free amphiphilic copolymer, P(4VPz-co-V4D4) with $F_{V4D4} = 67\%$, demonstrated similar deterrence of *C. lytica* biofilm, with a p -value of 0.1163. The antifouling performance of amphiphilic copolymers against *C. lytica* was thus attributed to its facile chain reorientation, which likely facilitates biofilm removal under shear forces in dynamic marine conditions.

To further assess biofouling caused by eukaryotes like diatoms, we extended our antifouling tests to *N. incerta*, a model diatom in marine biofouling studies and a known aggressive foulant. We compared the adhesion of *N. incerta* on P(4VPz-co-V4D4), $F_{V4D4} = 45\%$, P(4VPz-co-DVB) and P(4VPz-co-PFOA) against a commercial paint, i.e., Intersleek 1100SR (referred to as 1100SR), which is amphiphilic in nature and well-characterized to promote fouling release.^[65]

The surface-attached diatoms were quantified based on the fluorescence intensity of their chlorophyll, extracted after 2 h of incubation (Figure 6a, “before jetting”). To assess the fouling release properties, the diatom-fouled surfaces were subject to water-jetting at the preset pressures of 10 psi for 10 s, followed by chlorophyll extraction performed on the remaining surface-attached diatoms (Figure 6a, “10 psi”). The percentage of removal, calculated as the ratio of remaining to initial chlorophyll content, is shown in Figure 6b. Compared to 1100SR, P(4VPz-co-V4D4) reduced the initial diatom attachment by 18%, with a

p -value of 0.0993, indicating improved antifouling performance than a best-in-class commercial paint. In contrast, P(4VPz-co-DVB), and P(4VPz-co-PFOA), exhibited 39% and 31% higher initial attachment than 1100 SR, respectively. After water jetting at 10 psi, the remaining diatoms on the P(4VPz-co-V4D4) surface was merely 8% that on 1100SR (after the same treatment). This corresponds to a percentage removal of 96% at 10 psi on P(4VPz-co-V4D4), far exceeding the 60% removal observed for 1100SR. P(4VPz-co-DVB) and P(4VPz-co-PFOA) showed similar removal rates (96% and 97%, respectively) but retained 17% and 33% more diatoms than P(4VPz-co-V4D4).

When comparing the fouling results across prokaryotes and eukaryotes, P(4VPz-co-PFOA) and P(4VPz-co-V4D4) both showed leading antifouling effect against *P. aeruginosa* and *C. lytica*, whereas P(4VPz-co-V4D4) was most effective against *N. incerta*. This suggests that while molecular heterogeneities are effective against bacterial adhesion, the deterrence of eukaryotic foulers may require additional interfacial design, such as surface elasticity, and additional steps, such as water jetting, to achieve.

Notably, P(4VPz-co-V4D4) demonstrated superior deterrence/release of marine diatom fouling compared to zwitterionic P(4VPz-co-DVB) and amphiphilic P(4VPz-co-PFOA) (Figure 6a; Figure S7a, Supporting Information). Despite their comparable copolymer compositions, P(4VPz-co-V4D4) exhibited a surface diatom quantity that was 37% lower than P(4VPz-co-PFOA), which we attribute to the lower Young's modulus of the V4D4-based copolymer. As demonstrated in previous studies,^[66] a more elastic surface facilitates interfacial deformation under hydrodynamic shear, promoting foulant detachment. Compared to the zwitterionic coating, P(4VPz-co-DVB) ($F_{4VPz} = 87\%$), P(4VPz-co-V4D4) exhibited a 41% reduction in diatom attachment. While P(4VPz-co-DVB) exhibited lower Young's modulus than P(4VPz-co-PFOA), it remains higher than that of P(4VPz-co-V4D4), which, combined with the dynamic surface chain reorientation of the latter, may explain the reduced

diatom adhesion the V4D4-based copolymer. Moreover, AFM analysis confirmed that all tested coatings possess ultra-smooth surfaces, ruling out surface roughness as a differentiating factor in antifouling performance.

While low Young's modulus (like that for P(4VPz-co-V4D4)) is commonly associated with fouling release, it is worth noting that the diatom adhesion before water jetting showed the lowest adhesion on P(4VPz-co-V4D4) among all surfaces tested. Similarly, the commercial fouling release paint, Intersleek 1100SR, showed lower diatom adhesion than P(4VPz-co-DVB) and P(4VPz-co-PFOA). These observations hint at the correlation between the "before jetting" results and fouling-release capabilities. To test that correlation and further differentiate possibly subtle differences among the synthetic surfaces, we extended the *N. incerta* assay to a longer attachment period of 5 h and evaluated fouling release under milder conditions (water-jetting at 0.5 psi, Figure S7, Supporting Information). Under these conditions, P(4VPz-co-V4D4) demonstrated 51% and 58% less diatom attachment compared to P(4VPz-co-DVB) and P(4VPz-co-PFOA), respectively. After fouling removal at the low jetting pressure of 0.5 psi, the remaining diatoms on P(4VPz-co-V4D4) were 52% and 30% less than those on P(4VPz-co-DVB) and P(4VPz-co-PFOA), respectively. These results underscore P(4VPz-co-V4D4) coating's effectiveness in fouling release even under low-shear conditions that resemble regular sample rinsing and handling while confirming the importance of Young's modulus in enabling fouling release.

These results demonstrate that the fluorine-free amphiphilic copolymer P(4VPz-co-V4D4) outperforms conventional zwitterionic polymers, PFAS-based amphiphilic coatings, and commercial fouling-release paints in both antifouling and fouling-release properties, highlighting its promise as a sustainable solution in marine antifouling applications.

3. Conclusions

In this work, we introduced a new class of fluorine-free amphiphilic copolymers for the deterrence of marine biofouling, enabled by an all-dry synthesis method, iCVD. This new class of amphiphilic coatings features V4D4, which simultaneously serves as the hydrophobic component and an elastic crosslinker. The benefit of V4D4 is threefold: (i) It replaces the hydrophobic monomers in previous best-in-class amphiphilic copolymers, which invariably rely on fluorinated side chains to achieve amphiphilicity albeit causing environmental concerns with potential generation and release of PFAS; the new class of amphiphilic coatings reported here achieved unprecedented antifouling performance using fluorine-free chemistry, mitigating the environmental concerns associated with the deployment of this important class of antifouling materials; (ii) it affords excellent chemical stability, remaining stable even in highly caustic conditions, pointing to long-term applications, e.g., in marine transportation, with sufficient stability to withstand aggressive cleaning procedures; (iii) its siloxane bonds afford excellent chain flexibility, enabling an elastic interface that facilitated fouling release, whose performance surpassed a commercial amphiphilic fouling-release paint, namely Intersleek 1100SR. Additionally, the incorporation of V4D4 significantly increases the derivatization depth of the polymer film, ensuring a deeper and more complete conversion

of zwitterionic moieties, pointing to further prolonged antifouling effects in the unlikely event of coating degradation. This combination of properties is unprecedented in previous antifouling coatings. It addresses several critical gaps in the literature, namely the need to eliminate PFAS in coating design and the need to bridge prokaryotic fouling, e.g., by bacteria, and eukaryotic fouling, e.g., by diatoms. The simultaneous antifouling and fouling-release properties were demonstrated using a variety of foulants, including bacteria and marine diatoms, in both static and dynamic conditions. The successful replacement of PFAS with V4D4 while enabling a broader-spectrum efficacy in biofouling deterrence positions this new class of amphiphilic copolymers as a promising solution to sustainable antifouling technologies. While we focus on amphiphilic antifouling coatings in this study, other non-amphiphilic copolymers also offer fluorine-free solutions to the issue of biofouling, albeit via distinct mechanisms. Future research into a comprehensive comparison of different antifouling chemistries across distinct categories could offer valuable insight into the design of next-generation sustainable antifouling materials.

4. Experimental Section

Chemicals and Materials: The monomer 4-vinylpyridine (4VP, $\geq 95\%$), divinylbenzene (DVB, $\geq 80\%$), and initiator *tert*-butyl peroxide (TBPO, $\geq 98\%$) were acquired from Sigma-Aldrich (St. Louis, MO). The monomer 1H,1H,2H,2H-perfluorooctyl acrylate (PFOA, 97%) was obtained from Synquest Laboratories (Alachua, FL). The monomer 1,3,5,7-tetramethyl-1,3,5,7-tetravinyl cyclo-tetrasiloxane (V4D4, $\geq 97\%$) was acquired from AmBeed (Arlington Heights, IL). The derivatization molecule, 1,3-propane sultone (PS, $\geq 98\%$), was acquired from Sigma-Aldrich (St. Louis, MO). The 18 $\times$ 18 mm glass coverslips (12542A) were purchased from Fisher Scientific (Pittsburgh, PA). The silicon (Si) wafer substrates were acquired from Pure Wafer (Prescott, AZ). For biofilm formation experiments, crystal violet (CV) from Ward's Science (St. Catharines, ON, Canada), acetic acid (AA, 100%) from Merck (Darmstadt, Germany), and LB medium from BD Life Sciences (Sparks, MD) were used. All chemicals were used as purchased without further purification. A Milli-Q unit (Millipore) was used to produce deionized water with a resistivity of 18.2 M Ω -cm at 25 $^{\circ}$ C for all experiments.

iCVD Synthesis: The copolymers and their corresponding homopolymers were deposited in a custom-built reactor with a 2.5 cm thick glass lid, which allowed in situ laser interferometry (He – Ne laser, JDSU, USA) for visual observation and coating thickness control. Reactor pressure was maintained at a specific set point using a throttle butterfly valve (MKS Instruments, USA) and monitored by a capacitance manometer (Baratron, MKD Instruments, USA). Temperature measurements were taken using type K thermocouples (Omega Engineering, USA). The initiator, TBPO, was thermally decomposed by a filament array made of copper/nickel wire (2.8 cm above the stage) that was resistively heated by a DC power supply (B&K Precision, USA). The filament temperature was also measured by a type K thermocouple. The flow rates for the V4D4-containing copolymers are summarized in Table 1. The reactor pressure was maintained at 250 mTorr. The monomers were introduced into the reactor through needle valves (Swagelok, USA), with the monomer jar temperatures set to 50 $^{\circ}$ C for 4VP and 70 $^{\circ}$ C for V4D4. The initiator TBPO was kept at room temperature and fed into the reactor at a flow rate of 1 sccm using a mass flow controller (MKS Instruments, USA). The filament temperature was maintained at 250 $^{\circ}$ C, while the stage temperature was kept at 36 ± 1 $^{\circ}$ C. The monomer feed gas fraction of V4D4 (f_{V4D4}) was calculated based on its fractional saturation pressure (P_m/P_{sat}) divided by the total P_m/P_{sat} of all monomers used in a single deposition. The copolymer thin film coatings were maintained at a thickness of 100 nm on both sides of the coverslips. The molar fraction of V4D4 (F_{V4D4}) in the copolymer differed from f_{V4D4}

Table 1. Flow rates for P(4VP-co-V4D4) depositions. The substrate temperature was kept at 36 ± 1 °C, and chamber pressure was kept at 250 mTorr. f_{V4D4} represents the V4D4 monomer molar fraction on the substrate surface during the iCVD reaction, and F_{V4D4} represents the molar fraction in the thin film (measured via FTIR).^[25]

4VP flow rate [sccm]	V4D4 flow rate [sccm]	TBPO flow rate [sccm]	f_{V4D4} [%]	F_{V4D4} [%]
1.0	0.3	1.0	78	67
1.7	0.3	1.0	67	45
2.0	0.2	1.0	57	31

due to the varying reactivities of each monomer, as detailed in a previous report.^[67]

For comparison, P(4VP-co-DVB) and P(4VP-co-DVB) with $F_{DVB} = 50\%$, P(4VP-co-PFOA), $F_{PFOA} = 50\%$ films were synthesized following previously reported methods, detailed deposition conditions are listed in Tables S2–S4 (Supporting Information), respectively.^[17,20] All deposited coatings were subsequently subjected to PS derivatization for conversion of zwitterionic moieties. Thin film of PPFOA (homopolymer) was deposited following the condition in Table S3 (Supporting Information) but replacing 4VP flow with argon at the same flow rate. Thin film of PV4D4 (homopolymer) was synthesized under the same conditions described in Table 1 with V4D4 flow rate of 0.4 sccm and replacing the 4VP flow by argon at the same flow rate.

Vapor-Phase Derivatization: The reaction mechanism for the subsequent vapor-phase derivatization of P(4VP-co-V4D4) copolymer is shown in Figure 1c. After iCVD deposition, the coated substrates were exposed to PS vapor to synthesize pyridinium-based sulfobetaine zwitterionic functional groups.^[68] The coated substrates were mounted on a crystallizing dish (VWR) with 1 g of PS and wrapped in aluminum foil tightly. The crystallizing dish was placed inside a vacuum oven (Columbia International) that was maintained at 25 Torr and 60 °C overnight to allow the PS vapor to react with the copolymer thin film. The derivatized copolymers are denoted as P(4VPz-co-V4D4).

Chemical Characterization of the Polymer Thin Films: The thickness of the thin film on flat Si wafer substrates was measured by a J. A. Woolam Alpha-SE spectroscopic ellipsometer at three incident angles of 65°, 70°, and 75° with a wavelength range from 315 to 718 nm. A Cauchy-Urbach model was used to fit the data. FTIR measurements were performed on a Nicolet iS50 (Thermo Fisher Scientific) spectrometer in transmission mode. A deuterated triglycine sulfate (DTGS) detector was used to collect the spectra over the range of 500–4000 cm^{-1} with a resolution of 4 cm^{-1} . The data was averaged over 128 scans to ensure a sufficient signal-to-noise ratio. All collected spectra were normalized by the polymer film thickness and baseline-corrected by subtracting a background spectrum of silicone (Si) wafer using OMNIC software. XPS was carried out with a Scienta Omicron ESCA 2SR spectrometer with a monochromatized Al K α X-ray source at a power of 150 W and a voltage of 12 kV. The charge neutralizer was set to a beam energy of 5 eV, focus voltage of 300 V, and emission of 15 μA . The survey spectra were collected over 0–1200 eV. N(1s), C(2s) and Si(2p) high-resolution spectra were collected over 392–412 eV, 278–303 eV, and 98–106 eV, respectively. The elemental composition in samples was determined by using the CasaXPS software with a Shirley background. The charging was corrected with the reference value of 284.8 eV, which is the binding energy for C-C and C-H bonds arising in the C(1s) high-resolution spectra.^[69] XPS Depth profiling was performed using an Ar⁺ cluster gun with a cluster size of 500, an ion energy of 600 eV, and a raster size of 0.7 mm. The X-ray beam was set to a 400-micron spot size for optimal resolution and signal collection. A flood gun was employed for charge neutralization. Each etching step lasted for 30 s, with a total of 7 steps. The etching depth was calibrated by the total thickness of the thin film.

Surface Morphology and Modulus Characterization: AFM was performed with an Asylum Cypher ES microscope in AM-FM Viscoelastic mapping mode with PPP-NCR cantilevers ($k \approx 26$ N/m, $f \approx 300$ kHz,

Nanosensors). To quantify surface roughness, 5×5 μm regions of copolymer thin films (coated onto Si wafers) were scanned with a frequency of 0.5 Hz. First eigenmode in the AM mode was driven with the amplitude (A_1) ≈ 100 nm, while the second eigenmode in the FM mode was driven with the frequency (f_2) ≈ 1.8 MHz ($A_2 \approx 1$ nm). Based on bimodal AFM theory, the AFM tip was described by a Hertz model with punch tip geometry and radius of 10 nm. Calibration of the cantilever parameters was performed through an AM-FM measurement on a polystyrene reference sample, where the modulus was assumed to be 3.0 GPa. Subsequently, Bimodal AM-FM imaging was conducted on respective samples.

Dynamic Water Contact Angle: To determine the contact angle hysteresis, dynamic WCA, i.e., advancing and receding contact angles, were measured using a Rame-Hart Model 500 goniometer at room temperature with Milli-Q water using the volume-addition method. Water droplets were dispensed onto a surface by using an automatic dispenser. The advancing contact angle was obtained as the plateau angle during the increase in drop size from 1 to 10 μL , while the receding angle was the angle at which the angle plateaued while the volume was reduced from 10 to 0 μL .

Chemical Stability: To evaluate chemical stability, 1 cm $\times$ 1 cm coated Si wafers were submerged in 5 mL 0.5 M H_2SO_4 solution at room temperature for 5 hours to test acidic resistance. For alkaline resistance, samples were submerged in 1 M NaOH solution under the same conditions. After immersion, FTIR measurements were conducted as previously described.

Cytotoxicity: P(4VPz-co-V4D4) and an aluminum surface, as a reference, were soaked in 0.1 M NaOH solutions for 5 h, followed by neutralization with 0.1 M HCl. The neutralized solutions were then used to culture *C. lytica* for 5 h, and the bacterial CFU counts were compared by spreading the *C. lytica* culture without and with the exposure onto 3 plates.

P. aeruginosa Fouling Experiment: *P. aeruginosa* (PAO1 strain) was used for measuring the antifouling performance of the copolymer thin films at the three-phase interface. Bacterial strains were preserved in glycerol at -80 °C prior to the fouling experiments. The procedure of the fouling experiment using *P. aeruginosa* has been previously reported.^[70] Briefly, PAO1 cells were streaked from frozen stock onto LB agar plates and incubated overnight (≈ 16 h) at 37 °C. A single colony was then inoculated into LB medium and cultured for 18 h at 37 °C and 225 rpm. Following incubation, the optical density at 600 nm (OD_{600}) reached ≈ 0.5 –0.6 before the culture was diluted 10 times for the fouling experiments. Coated glass coverslips, bare glass coverslip controls, custom-made aluminum discs controls (15 mm radius) and 12-well plates (3513, Corning Inc., Corning, NY) were sterilized under UV light for 1 h before the fouling experiments. 3 mL of the ten-fold diluted overnight cultures were added to each well of the 12-well plate. The lid, mounted with coverslips or aluminum discs, was placed on top of the well plate to maintain the vertical orientation of the coverslips in the medium, ensuring biofilm growth at the solid-liquid-air interface. The setup was then incubated for 8 h at 37 °C. The biomass quantification was performed using CV staining. Briefly, the coverslips were rinsed with Milli-Q water and then submerged into a 0.5% (w/v) CV solution for 15 min. After staining, the samples were air-dried overnight and subsequently submerged in 33% (v/v) AA for 15 min to dissolve the stained biofilm. The OD_{550} was measured for each sample, along with two control samples of AA for background correction. For aluminum controls, the stained biofilm area was imaged and analyzed using ImageJ. Since the contact areas of aluminum discs (circular) and glass coverslips (square) were different, the OD_{550} measured on aluminum was calibrated by the ratio of the stained biofilm area on aluminum to that on glass coverslips.

Custom-Made Flow Cell System for C. Lytica Fouling Experiment: A custom-made flow cell system was used to perform the fouling experiment of *C. lytica* under shearing conditions. *C. lytica* was selected as a model marine bacterium due to its wide distribution and relevance to marine fouling studies.^[71,72] The H11 strain of *C. lytica* was generously provided by Prof. Michael Hadfield at University of Hawai'i at Mānoa. *C. lytica* was streaked from frozen stock onto marine agar plates (5 g peptone, 1 g yeast extract, 15 g agar in 1 L artificial seawater (ASW, 30 g/L artificial sea salt from Instant Ocean®)) and let grow for 24 hours at room temperature (22 °C). A single colony was then inoculated into marine medium (5 g peptone, 1 g yeast extract in 1 L ASW) and cultured for 24 h at room temperature

and 160 rpm. After 24 h of incubation, the cell density reached $\approx 1.0 \times 10^8$ cells/mL before the culture was diluted tenfold for the fouling experiment.

The custom-made flow cell system included a custom-made hollow aluminum framework (24 mm $W \times$ 40 mm $L \times$ 8 mm D) serving as the flow cell chamber, clamped between layers of polydimethylsiloxane (PDMS) and a layer of micro slides (VWR) on each side. Three replicates of circle microscope cover glass (15 mm radius, NEST) or custom-made aluminum discs (15 mm radius)^[64] coated with desired chemistries were attached to one side of a PDMS roof. A high-precision peristaltic pump (ISMATEC®) controlled the flow of marine medium from a 4 L culture medium bottle into a 4 L waste bottle (ACCFL0008, ACCFL0009, IBI Scientific). Before starting the fouling experiment, the flow cell system was sterilized by flowing through 70% ethanol at a flow rate of 4 mL/h for 1 h, followed by washing with marine medium to remove any residual ethanol for 1 h. The cell suspension, as prepared earlier, was injected into the flow cells using a sterilized syringe with a volume at least three times that of the flow cells. In the first hour of the fouling experiment, the flow cells were positioned so that the samples were at the bottom, allowing sufficient cell adhesion onto the samples via cell sedimentation without medium flow. After the first hour, the flow cells were inverted to position the samples on top (to avoid non-specific fouling), with the marine medium flow set at a flow rate of 4 mL/h. This process continued for 24 h to ensure biofilm formation. After 24 h, the flow cells were disassembled, and the cover glass samples were gently retrieved and washed three times to remove loosely attached bacteria. The biomass quantification was performed using CV staining, as described in the previous subsection.

Evaluation of Antifouling and Fouling Release Performance Against the Diatom *N. incerta*: The method for evaluating antifouling performances of copolymer surfaces prepared in 24-well plates against marine diatoms have been previously described in detail.^[73] Briefly, five-day-old cultures of the marine brown microalgae diatom *N. incerta* were re-suspended in fresh Guillard's F/2 medium prepared in ASW to achieve a final cell density of 10^5 cells/mL. Subsequently, 1.0 mL of the resulting diatom suspension was added to each well of the coating plates and incubated at 18 °C for 2 h under light conditions (photon flux density 33 mmol m⁻² s⁻¹) to facilitate cell attachment. After the 2-hour cell attachment period elapsed, the plates were transferred to the deck of an automated water-jet apparatus and subjected to water-jet treatments at nozzle pressures of 10 psi for 10 s. The initially attached cells and remaining surface-attached cells after water-jet treatments were quantified by adding 0.6 mL of DMSO to each well for 20 min to extract all chlorophyll, and 0.15 mL of the resulting eluates was transferred to a 96-well plate to quantify the fluorescence of chlorophyll (Ex: 360 nm; Em: 670 nm) using a multi-well plate spectrophotometer. The percentage of diatom removal was calculated by comparing the total biomass on the coating surfaces before and after the water-jet treatments using the following formula:

$$\text{Percentage Diatom Removal} = \left(1 - \left(\phi_{\text{jetted}} / \phi_{\text{non-jetted}} \right) \right) \times 100 \quad (1)$$

where ϕ_{jetted} is the mean chlorophyll fluorescence value of jetted samples (in replicates of three) and $\phi_{\text{non-jetted}}$ is the mean chlorophyll fluorescence value of non-jetted samples (in replicates of three).^[74]

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

amphiphilic polymer, antifouling materials, biofouling, initiated chemical vapor deposition, marine fouling, PFAS

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